

Numerical solutions of Ordinary differential Equations

A number of problems in science & technology can be formulated into differential equations. The analytical methods of solving differential equations are applicable only to a limited type of eq. Quite often differential equations appearing in physical problems do not belong to any of these familiar types & one is obliged to resort to numerical methods. These methods are of even greater importance when we realise that computing machines are now readily available which reduce numerical work considerably.

Solution of a differential Equation - The solution of an ordinary differential equation means finding an explicit expression for y in terms of a finite number of elementary functions of x . Such a solution of a differential equation is known as the closed or finite form of solution. In the absence of such a solution, we have recourse to numerical methods of solution.

Let us consider the first order differential equation

$$\frac{dy}{dx} = f(x, y) ; \text{ given } y(x_0) = y_0$$

to study the various numerical methods of solving such equations. In most of these methods, we replace the differential equation by a difference equation & then solve it. These methods yield solutions either as a power series in x from which the values of y can be found by direct substitution or a set of values of x & y . The Picard's method belong to the former class of solutions. These methods are referred to as single step methods. The methods of Euler, Runge-Kutta & Milne predictor belong to the latter class of solutions. These methods are also known as step-by-step methods.

Initial & boundary conditions - An ordinary differential equation of the n th order is of the form

$$F(x, y, dy/dx, d^2y/dx^2, \dots, d^ny/dx^n) = 0$$

Its general solution contains n arbitrary constants & is of the form

$$\phi(x, y, C_1, C_2, \dots, C_n) = 0$$

To obtain its particular solution, n conditions must be given so that the constants $C_1, C_2, \dots, C_n$ can be determined. If these conditions are prescribed at one point (say x_0) then the differential equation together with the condition constitute an initial value problem of n th order. If the conditions are prescribed at two or more points, then the problem is termed as boundary value problem. We will only consider initial value problems here.

Picard's Method—

Consider the first order equation

$$\frac{dy}{dx} = f(x, y) \quad \text{--- (1)}$$

It is required to find that particular solution of (1) which assumes the value y_0 when $x = x_0$.

Integrating (1) between limits, we get

$$\int_{y_0}^y dy = \int_{x_0}^x f(x, y) dx$$

$$y = y_0 + \int_{x_0}^x f(x, y) dx. \quad \text{--- (2)}$$

This is an integral equation equivalent to (1), for it contains the unknown y under the integral sign.

As a first approximation y_1 to the solution, we put $y = y_0$ in $f(x, y)$ & integrating (2), we get

$$y_1 = y_0 + \int_{x_0}^x f(x, y_0) dx.$$

For a second approximation y_2 , we put $y = y_1$ in $f(x, y)$

Integrate ②, giving

$$y_2 = y_0 + \int_{x_0}^x f(x, y_1) dx.$$

Similarly third approximation is

$$y_3 = y_0 + \int_{x_0}^x f(x, y_2) dx$$

Continuing, this process, we obtain $y_4, y_5, \dots, y_n$ where

$$y_n = y_0 + \int_{x_0}^x f(x, y_{n-1}) dx.$$

Hence this method gives a sequence of approximations $y_1, y_2, \dots, y_n$ each giving a better result than the preceding one.

Note: This can be applied only to a limited class of eqs in which the successive integrations can be performed easily.

e.g: Using Picard's method, obtain a solution upto fifth approximation of the equation $dy/dx = y+x$, such that $y=1$ when $x=0$.

solu: we have $y = 1 + \int_0^x f(x, y) dx$

$$y = 1 + \int_0^x (x+y) dx$$

First approx₁: Put $y=1$ in $(y+x)$, we get

$$y_1 = 1 + \int_0^x (1+x) dx = 1 + \left[x + \frac{x^2}{2} \right]_0^x = 1 + x + \frac{x^2}{2}.$$

second approx: Put $y = 1 + x + \frac{x^2}{2}$ in $(y+x)$, we get

$$\begin{aligned} y_2 &= 1 + \int_0^x \left(1 + x + \frac{x^2}{2} + x \right) dx \\ &= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^2}{2} = 1 + x + x^2 + \frac{x^3}{6}. \end{aligned}$$

third approx: Put $y = 1 + x + x^2 + \frac{x^3}{6}$ in $(y+x)$, we get

$$\begin{aligned}
 y_3 &= 1 + \int_0^x \left(1 + x + x^2 + \frac{x^3}{6} + x\right) dx \\
 &= 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{24} + \frac{x^2}{2} \\
 &= 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{24}
 \end{aligned}$$

Fourth approx: Putting $y = y_3$; we get

$$\begin{aligned}
 y_4 &= 1 + \int_0^x \left(1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{24} + x\right) dx \\
 &= 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{120} + \frac{x^2}{2} \\
 &= 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{120}
 \end{aligned}$$

Fifth approx: Putting $y = y_4$ in $(y+x)$, we get

$$\begin{aligned}
 y_5 &= 1 + \int_0^x \left(1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{120} + x\right) dx \\
 &= 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{60} + \frac{x^6}{720} + \frac{x^2}{2} \\
 &= 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{60} + \frac{x^6}{720} \quad \text{is the soln.}
 \end{aligned}$$

e.g: Find an approximate value of y when $x = 0.1$, if $dy/dx = x - y^2$ and $y = 1$ at $x = 0$ using Picard's method.

Solu:

$$y = 1 + \int_0^x (x - y^2) dx$$

First approx: Put $y = 1$, we get

$$y_1 = 1 + \int_0^x (x - 1) dx = 1 + \frac{x^2}{2} - x = 1 - x + \frac{x^2}{2}$$

second approx : Put $y = 1 - x + \frac{x^2}{2}$, we get

$$\begin{aligned} y_2 &= 1 + \int_0^x \left[x - \left[1 - x + \frac{x^2}{2} \right] \right] dx \\ &= 1 + \int_0^x x - \left(1 + x^2 + \frac{x^4}{4} - 2x - x^3 + x^2 \right) dx \\ &= 1 + \int_0^x \left(-1 + 3x + 2x^2 + (-x^3) + \frac{x^4}{4} \right) dx \\ &= 1 - x + 3\frac{x^2}{2} + \frac{2x^3}{3} - \frac{x^4}{4} + \frac{x^5}{20} \end{aligned}$$

$$y_2|_{0.1} = 0.9155$$

Ans: 0.9138.

e.g: Use picard's method to solve $dy/dx = 2x - y$ given that $y = 0.9$ at $x = 0$. Find value of y at $x = 0.2, 0.4, 0.6$.

Ans: 0.7743, 0.7439, 0.7915.

Euler's Method -

This method is very elementary and should not be used for practical solutions. However, the principle involved is of vital importance. More accurate methods which can be used in practice are given later.

Let $x_0, x_1, x_2, \dots$ be equispaced values of x at a spacing h and let $y_0, y_1, y_2, \dots$ be the corresponding values of y on a curve satisfying the differential eq.

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \quad \text{--- ①}$$

Suppose the solution of ① is $y = \phi(x)$.

For the first step we find the value of y at $x = x_0 + h$, & call it y_1 .

$$\begin{aligned}
 \text{then } y_1 &= \phi(x_1) = \phi(x_0 + h) \\
 &= \phi(x_0) + h\phi'(x_0) + \frac{h^2}{2!}\phi''(x_0) + \dots \\
 &= y_0 + hf(x_0, y_0) + \frac{h^2}{2!}f'(x_0, y_0) \quad (\text{using ①})
 \end{aligned}$$

Hence if h is small, neglecting terms involving h^2 & higher powers of h , we have

$$y_1 = y_0 + hf(x_0, y_0) \quad \text{--- ②}$$

On the same lines, for the second step, we take (x_1, y_1) as the starting point and get

$$y_2 = y_1 + hf(x_1, y_1)$$

Continuing the process, we have in general

$$y_{n+1} = y_n + hf(x_n, y_n) \quad \text{--- ③}$$

This formula can be used to determine y_{n+1} when y_n is known. Putting in ③, $n=0, 1, 2, \dots$ we can successively determine $y_1, y_2, y_3, \dots$ provided the starting point (x_0, y_0) is known.

eg: Applying Euler's method, solve

$$\frac{dy}{dx} = x+y; \quad y(0)=0$$

choosing the step length $= 0.2$. Carry out 6 steps.

solu: we take $n=5$; $h=0.2$. The calculations are arranged as follows:

x	y	$x+y = dy/dx$	$\text{old } y + 0.2(dy/dx) = \text{new}$
0.0	0.0	0.0	$0.0 + 0.2 \times 0.0 = 0.0$
0.2	0.0	0.2	$0.0 + 0.2 \times 0.2 = 0.04$
0.4	0.04	$0.4 + 0.04 = 0.44$	$0.04 + 0.2 \times 0.44 = 0.128$
0.6	0.128	$0.6 + 0.128 = 0.728$	$0.128 + 0.2 \times 0.728 = 0.2736$
0.8	0.2736	$0.8 + 0.2736 = 1.0736$	$0.2736 + 0.2 \times 1.0736 = 0.48832$
1.0	0.48832	$1.0 + 0.48832 = 1.48832$	$0.48832 + 0.2 \times 1.48832 = 0.785$

$$1.859$$

$$1.2 + 0.7859 = 1.9859$$

$$0.7859 + 0.2 \times 1.9859$$

$$= 1.18308$$

Ans:

Q : Using Euler's method, find an approx value of y corresponding to $x=1$, given that $dy/dx = x+y$ & $y=1$ when $x=0$.

Solu : we take $n=10$, $h=0.1$. The various calculations are as follows:

x	y	$x+y = dy/dx$	old $y + 0.1 \times dy/dx = \text{New } y$
0.0	1.00	1.00	$1.00 + 0.1 \times (1.00) = 1.10$
0.1	1.10	1.20	$1.10 + 0.1 \times (1.20) = 1.22$
0.2	1.22	1.42	$1.22 + 0.1 \times (1.42) = 1.36$
0.3	1.36	1.66	$1.36 + 0.1 \times (1.66) = 1.53$
0.4	1.53	1.93	$1.53 + 0.1 \times (1.93) = 1.72$
0.5	1.72	2.22	$1.72 + 0.1 \times (2.22) = 1.94$
0.6	1.94	2.54	$1.94 + 0.1 \times (2.54) = 2.19$
0.7	2.19	2.89	$2.19 + 0.1 \times (2.89) = 2.48$
0.8	2.48	3.29	$2.48 + 0.1 \times (3.29) = 2.81$
0.9	2.81	3.71	$2.81 + 0.1 \times (3.71) = 3.18$
1.0	3.18		

$\therefore$ the req. ⁴ approx value of $y = 3.18$.

Modified Euler's Method - According to modified Euler's method we find a better approximation of y_1 as $y_1^{(1)}$ by taking

$$y_1^{(1)} = y_0 + h \left[\frac{f(x_0, y_0) + f(x_0+h, y_1)}{2} \right] \quad \text{--- ①}$$

So the eq. $y_1 = y_0 + hf(x_0, y_0)$ is called the predictor whereas eq ① is called corrector of y_1 . Now insert $y_1^{(1)}$ as

y_1 in ① as we get a still better value of y_1 as $y_1^{(2)}$ by

$$y_1^{(2)} = y_0 + h \left[\frac{f(x_0, y_0) + f(x_0 + h, y_1^{(1)})}{2} \right]$$

The process is repeated until two consecutive values of y agree. Once y_1 is obtained to the desired degree of accuracy, y corresponding to $(x_0 + 2h)$ is found from predictor

$$y_2 = y_1 + h f(x_0 + h, y_1)$$

& a better approximation $y_2^{(1)}$ is obtained by corrector

$$y_2^{(1)} = y_1 + h \left[\frac{f(x_0 + h, y_1) + f(x_0 + 2h, y_2)}{2} \right]$$

The step is repeated till we get y_2 correct to desired degree of accuracy. Next we compute y_3 as above & so on.

This predictor-corrector method is of great importance in practical problems.

e.g.: Using modified Euler's method, find an approximate value of y when $x = 0.2$ given that $dy/dx = x + y$ & $y = 1$ when $x = 0$.

Solu: To compute $y(0.2)$ we have $x_0 = 0$, $y_0 = 1$, $h = 0.1$.

$$f(x, y) = x + y \quad ; \quad f(x_0, y_0) = 0 + 1 = 1$$

By predictor formula

$$\begin{aligned} y_1 &= y_0 + h f(x_0, y_0) \\ &= 1 + 0.1 \times [0 + 1] = 1.1 \end{aligned}$$

Then applying corrector formula

$$\begin{aligned} y_1^{(1)} &= y_0 + h \left[\frac{f(x_0, y_0) + f(x_0 + h, y_1)}{2} \right] \\ &= 1 + 0.1 \left[\frac{1 + (0.1 + 1.1)}{2} \right] = 1 + \frac{0.1 \times 2.2}{2} = 1 + 0.11 \\ &= 1.11 \end{aligned}$$

again applying corrector

(5)

$$y_1^{(2)} = y_0 + h \left[\frac{f(x_0, y_0) + f(x_0 + h, y_1^{(1)})}{2} \right]$$
$$= 1 + 0.1 \left[\frac{1 + (0.1 + 1.11)}{2} \right] = 1 + 0.1 \times \frac{2.21}{2} = 1.1105$$

Again applying corrector

$$y_1^{(3)} = y_0 + h \left[\frac{f(x_0, y_0) + f(x_0 + h, y_1^{(2)})}{2} \right]$$
$$= 1 + 0.1 \left[\frac{1 + (0.1 + 1.1105)}{2} \right] = 1 + 0.1 \times \frac{2.2105}{2} = 1.1105$$

Since $y_1^{(2)}$ & $y_1^{(3)}$ are equal upto third place at least we have
 $y(0.1) = 1.1105$.

Now we compute $y(0.2)$. By predictor formula

$$y_1 = y_0 + hf(x_0, y_0)$$

Here we take $x_0 = 0.1$; $h = 0.1$, $y_0 = 1.1105$,

$$f(x_0, y_0) = 0.1 + 1.1105 = 1.2105.$$

$$\therefore y_1 = 1.1105 + 0.1 \times 1.2105 = 1.2316$$

Using corrector.

$$y_1^{(1)} = y_0 + h \left[\frac{f(x_0, y_0) + f(x_0 + h, y_1)}{2} \right]$$
$$= 1.1105 + 0.1 \left[\frac{1.2105 + (0.2 + 1.2316)}{2} \right]$$
$$= 1.2426$$

$$y_1^{(2)} = y_0 + h \left[\frac{f(x_0, y_0) + f(x_0 + h, y_1^{(1)})}{2} \right]$$
$$= 1.1105 + 0.1 \left[\frac{1.2105 + (0.2 + 1.2426)}{2} \right]$$
$$= 1.2432$$

$$y_{n+1}^{(3)} = y_0 + h \left[\frac{f(x_0, y_0) + f(x_0+h, y_1^{(1)})}{2} \right]$$

$$= 1.1105 + 0.1 \left[\frac{1.2105 + (0.2 + 1.2432)}{2} \right]$$

$$= 1.2432$$

So $y(0.2) = 1.2432$ approx.

By we can find $y(0.3) = 1.4004$ approx.

Runge Kutta's method of fourth order —

This method do not require the calculations of higher order derivatives and give greater accuracy. This method possess the advantage of requiring only the function values at some selected points. This method is most commonly used & is often referred to as Runge-Kutta method only.

Working rule for finding the increment k of y corresponding to an increment h of x by Runge Kutta method is as follows —

$$\frac{dy}{dx} = f(x, y); \quad y(x_0) = y_0$$

is as follows —

calculate successively $k_1 = hf(x_0, y_0)$

$$k_2 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1)$$

$$k_3 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2)$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

& finally compute

$$K = \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

which gives the required approximate value as

$$y = y_0 + K.$$

One of the advantages of these methods is that the operation is identical whether the differential equation is linear or non linear. ⑥

e.g: Apply Runge-Kutta method (fourth order) to find an approximate value of y when $x=0.2$ given that $dy/dx = x+y$ and $y=1$, when $x=0$
Solu:

$$\frac{dy}{dx} = x+y \quad y(0)=1.$$

$$x_0 = 0, y_0 = 1, f(x, y) = x+y, h = 0.2$$

$$k_1 = h f(x_0, y_0) = 0.2 [0+1] = 0.2$$

$$\begin{aligned} k_2 &= h f\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right) \\ &= 0.2 \left[\left(0 + \frac{1}{2} \cdot 0.2\right) + \left(1 + \frac{1}{2} \cdot 0.2\right) \right] \\ &= 0.2 [0.1 + 1.1] = 0.24. \end{aligned}$$

$$\begin{aligned} k_3 &= h \left[f\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right) \right] \\ &= 0.2 \left[\left(0 + \frac{1}{2} \cdot 0.2\right) + \left(1 + \frac{1}{2} \cdot 0.24\right) \right] \\ &= 0.2 [0.1 + 1.12] = 0.2 \times 1.22 = 0.244. \end{aligned}$$

$$\begin{aligned} k_4 &= h \left[f(x_0 + h, y_0 + k_3) \right] \\ &= 0.2 [(0 + 0.2) + (1 + 0.244)] \\ &= 0.2 [1.444] = 0.2888. \end{aligned}$$

$$\begin{aligned} \text{Finally } R &= \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \\ &= \frac{1}{6} [0.2 + 2 \times 0.24 + 2 \times 0.244 + 0.2888] \\ &= \frac{1}{6} [0.2 + 0.48 + 0.488 + 0.2888] \\ &= 0.2428. \end{aligned}$$

$$\therefore y(0.2) = y_0 + R = 1 + 0.2428 = 1.2428.$$

e.g : Use Runge-Kutta's method to approximate y when x
given $y(1) = 1.2$ & $dy/dx = 3x + y^2$.

Solu : Given $\frac{dy}{dx} = 3x + y^2$; $x_0 = 1$, $y_0 = 1.2$; $h = 0.1$; $f(x, y) = 3x + y^2$

$$\text{Now } k_1 = hf(x_0, y_0)$$

$$= 0.1 [3(1) + (1.2)^2] = 0.1 [3 + 1.44] = 0.444.$$

$$k_2 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1)$$

$$= 0.1 [3(x_0 + \frac{1}{2}h) + (y_0 + \frac{1}{2}k_1)^2]$$

$$= 0.1 [3(1 + \frac{1}{2}0.1) + (1.2 + \frac{1}{2}0.444)^2]$$

$$= 0.1 [3.15 + 2.022] = 0.5172.$$

$$k_3 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2)$$

$$= 0.1 [3(x_0 + \frac{1}{2}h) + (y_0 + \frac{1}{2}k_2)^2]$$

$$= 0.1 [3.15 + (1.2 + \frac{1}{2}0.5172)^2]$$

$$= 0.1 [3.15 + 2.1275] = 0.5277$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

$$= 0.1 [3(x_0 + h) + (y_0 + k_3)^2]$$

$$= 0.1 [3(1 + 0.1) + (1.2 + 0.5277)^2]$$

$$= 0.1 [3.3 + 2.9849] = 0.6284.$$

$$\text{Now } K = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$= \frac{1}{6} [0.444 + 2 \times 0.5172 + 2 \times 0.5277 + 0.6284]$$

$$= \frac{1}{6} \times 2.1068 = \cancel{0.3511} 0.5270$$

$$\therefore y(1.1) = y_0 + K = 1.2 + \cancel{0.3511} 0.5270$$

$$= \cancel{1.5511} 1.7270.$$

Taylor Series Method—

Consider the first order equation

$$\frac{dy}{dx} = f(x, y) \quad \text{--- (1)}$$

Differentiating (1), we have

$$\frac{d^2y}{dx^2} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} \quad \text{--- (2)}$$

$$\text{i.e. } y'' = f_x + f_y f'$$

Differentiating this successively, we can get y''' , $y^{(iv)}$ etc. Putting $x = x_0$ & $y = y_0$ the values of $y'(0)$, $y''(0)$, $y'''(0)$ etc can be obtained. Hence the Taylor's series

$$y = y_0 + (x - x_0)(y')_0 + \frac{(x - x_0)^2}{2!}(y'')_0 + \frac{(x - x_0)^3}{3!}(y''')_0 + \dots \quad \text{--- (3)}$$

gives the value of y for every value of x for which (3) converges.

Note: This is a single step method but is useful only if the derivatives of higher orders can be calculated easily. If $f(x, y)$ is complicated & the calculation of higher orders derivatives becomes difficult then this method is ^{not} much useful. This is the main drawback of this method. However it can be used to find starting values for the calculation & application of powerful methods like Runge Kutta method, Milne predictor method etc which we will discuss later.

eg: Find an approximate value of y when $x = 0.1$ if $\frac{dy}{dx} = x - y^2$ & $y = 1$ using Taylor's method.

Solu: Since $\frac{dy}{dx} = x - y^2$

$$y' = x - y^2$$

$$y'' = 1 - 2yy'$$

$$y''' = 0 - 2y'y' - 2yy'' = -2(y')^2 - 2yy''$$

$$y^{iv} = -4y'y'' - 2y'y'' - 2yy'''$$

$$= -6y'y'' - 2yy'''$$

At $x=0$; $y=1$ i.e $y(0)=1$

$$y'(0) = -1$$

$$y''(0) = 3$$

$$y'''(0) = -8$$

$$y^{iv}(0) = 34$$

Putting these values in the Taylor's series, we have.

$$y = \cancel{1} y(0) + (x-0)y'(0) + \frac{(x-0)^2}{2!} y''(0) + \frac{(x-0)^3}{3!} y'''(0) + \frac{(x-0)^4}{4!} y^{iv}(0) + \dots$$

$$= 1 + x(-1) + \frac{x^2}{2!}(3) + \frac{x^3}{3!}(-8) + \frac{x^4}{4!}(34) + \dots$$

$$\begin{aligned} \text{So } y(0.1) &= 1 + (0.1)(-1) + \frac{(0.1)^2}{2!}(3) + \frac{(0.1)^3}{3!}(-8) + \frac{(0.1)^4}{4!}(34) + \dots \\ &= 1 - 0.1 + 0.015 - 0.00133 + 0.000141 \\ &= 0.913 \end{aligned}$$

eg: Find the value of y at $x=0.1$ and $x=0.2$ to five places of decimal from $\frac{dy}{dx} = x^2y - 1$; $y(0)=1$

Solu: $\frac{dy}{dx} = x^2y - 1$

$$y' = x^2y - 1$$

$$y'(0) = -1$$

$$y'' = 2xy + x^2y' \quad y''(0) = 0$$

$$y''' = 2y + 4xy' + x^2y'' \quad y'''(0) = 2$$

$$y^{iv} = 6y' + 6xy'' + x^2y''' \quad y^{iv}(0) = -6$$

Putting these in Taylor series

$$\begin{aligned} y(x) &= y(0) + (x-0)y'(0) + \frac{(x-0)^2}{2!}y''(0) + \dots \\ &= 1 + x(-1) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(2) + \frac{x^4}{4!}(-6) + \dots \end{aligned}$$

$$\text{Now } y(x) = 1 - x + \frac{2x^3}{3!} - \frac{6x^4}{4!} - \dots$$

$$y(0.1) = 0.90033$$

$$y(0.2) = 0.80227.$$

Runge's Predictor-Corrector Method:

$$\text{Let } \frac{dy}{dx} = f(x, y)$$

$$y(x_0) = y_0 \text{ is given.}$$

We first compute

$$y_1 = y(x_0 + h)$$

$$y_2 = y(x_0 + 2h)$$

$$\& y_3 = y(x_0 + 3h) \text{ by Picard's method.}$$

Next we calculate

$$f_0 = f(x_0, y_0)$$

$$f_1 = f(x_0 + h, y_1)$$

$$f_2 = f(x_0 + 2h, y_2)$$

$$f_3 = f(x_0 + 3h, y_3)$$

(This we will do by putting $x = x_0, x_0 + h, x_0 + 2h$ & $x_0 + 3h$ in place of x & $y = y_0, y_1, y_2, y_3$ in place of y , respectively in $f(x, y)$).

Then we find $y_4 = y(x_0 + 4h)$ by using

$$y_4 = y_0 + \frac{4h}{3} (2f_1 - f_2 + 2f_3)$$

which is called predictor.

Next we obtain

$$f_4 = f(x_0 + 4h, y_4)$$

Then a better value of y_4 is found by using

$$y_4 = y_2 + \frac{h}{3} (f_2 + 4f_3 + f_4)$$

which is called corrector.

Then an improved value of f_4 is computed & again the corrector is applied to find a still better value of y_4 . we repeat this step until y_4 remains unchanged.

Once y_4 & f_4 are obtained to desired degree of accuracy y_5 is found from predictor as

$$y_5 = y_1 + \frac{4h}{3} (2f_2 - 4f_3 + 2f_4)$$

& $f_5 = f(x_0 + 5h, y_5)$ is calculated.

Then a better approximation of the values of y_5 is obtained from corrector as

$$y_5 = y_3 + \frac{h}{3} (f_3 + 4f_4 + f_5)$$

We repeat this step till y_5 becomes stationary & then we proceed to calculate y_6 as before.

eg: Apply Milne's predictor corrector method to find a solution of the differential equation

$$\frac{dy}{dx} = x - y^2 \text{ in the range } 0 \leq x \leq 1 \text{ for the boundary}$$

condition $y=0$ at $x=0$. $h=0.2$

Solu: Using Picard's method

$$y = y(0) + \int_0^x f(x, y) dx \text{ when } f(x, y) = x - y^2$$

To get the first approx, we put $y=0$ in $f(x, y)$

$$y_1 = 0 + \int_0^x x dx = \frac{x^2}{2}$$

For second approx, we put $y=y_1$ in $f(x, y)$

$$y_2 = 0 + \int_0^x \left[x - \left(\frac{x^2}{2} \right)^2 \right] dx = \frac{x^2}{2} - \frac{x^5}{20}$$

For third approx, we put $y=y_2$ in $f(x, y)$

$$\begin{aligned} y_3 &= 0 + \int_0^x \left[x - \left(\frac{x^2}{2} - \frac{x^5}{20} \right)^2 \right] dx \\ &= \frac{x^2}{2} - \frac{x^5}{20} + \frac{x^8}{160} - \frac{x^{11}}{4400} \end{aligned} \quad \text{--- (1)}$$

Now let us determine the starting values of the Milne's method from (1) by choosing $h=0.2$.

$$x_0 = 0.0 \quad y_0 = 0.0 \quad f_0 = 0.0000$$

$$x_1 = 0.2 \quad y_1 = 0.020 \quad f_1 = 0.1996$$

$$x_2 = 0.4$$

$$y_2 = 0.0795$$

$$f_2 = 0.3937$$

$$x_3 = 0.6$$

$$y_3 = 0.1762$$

$$f_3 = 0.5689$$

using the predictor,

$$y_4 = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3) \\ = 0.3049$$

$\therefore$

$$x_4 = 0.8$$

$$y_4 = 0.3049$$

$$f_4 = 0.7070$$

And the corrector

$$y_4 = y_2 + \frac{h}{3}[f_2 + 4f_3 + f_4] \text{ yields}$$

$$y_4 = 0.3046, f_4 = 0.7072$$

Again using corrector

$$y_4 = 0.3046$$

since value of y_4 is stationary $\therefore$ we now jump to y_5 .

Now using the predictor

$$y_5 = y_1 + \frac{4h}{3}(2f_2 - f_3 + 2f_4) = 0.4554$$

$$x = 1.0$$

$$y_5 = 0.4554$$

$$f_5 = 0.7926$$

and the corrector

$$y_5 = y_3 + \frac{h}{3}(f_3 + 4f_4 + f_5), \text{ gives}$$

$$y_5 = 0.4555$$

$$f_5 = 0.7925$$

Again using corrector

$$y_5 = 0.4555, \text{ a value which is the same as last one}$$

$\therefore$

$$y(1) = 0.4555: (\text{Ans})$$

Adam Bashforth method —

Given $\frac{dy}{dx} = f(x, y)$ & $y_0 = y(x_0)$, we compute

$$y_{-1} = y(x_0 - h)$$

$$y_{-2} = y(x_0 - 2h)$$

$$y_{-3} = y(x_0 - 3h)$$

by Taylor's series or Euler's or Runge Kutta method.

Next we calculate

$$f_{-1} = f(x_0 - h, y_{-1})$$

$$f_{-2} = f(x_0 - 2h, y_{-2})$$

$$f_{-3} = f(x_0 - 3h, y_{-3})$$

Then to find y_1 , we substitute Newton's backward Interpolation formula;

$$f(x, y) = f_0 + n \nabla f_0 + \frac{n(n+1)}{2!} \nabla^2 f_0 + \frac{(n)(n+1)(n+2)}{3!} \nabla^3 f_0 + \dots$$

$$\text{in } y_1 = y_0 + \int_{x_0}^{x_0+h} f(x, y) dx \quad (1)$$

$$y_1 = y_0 + \int_{x_0}^{x_0+h} \left[f_0 + n \nabla f_0 + \frac{n(n+1)}{2!} \nabla^2 f_0 + \dots \right] dx$$

$$\text{Put } x = x_0 + nh$$

$$dx = h dn$$

$$= y_0 + h \int_0^1 \left[f_0 + n \nabla f_0 + \frac{n(n+1)}{2!} \nabla^2 f_0 + \dots \right] dn$$

$$= y_0 + h \left[f_0 + \frac{1}{2} \nabla f_0 + \frac{5}{12} \nabla^2 f_0 + \frac{3}{8} \nabla^3 f_0 + \dots \right]$$

Neglecting fourth & higher order differences & expressing ∇f_0 , $\nabla^2 f_0$ & $\nabla^3 f_0$ in terms of function values, we get

$$y_1 = y_0 + \frac{h}{24} (55f_0 - 59f_{-1} + 37f_{-2} - 9f_{-3}) \quad (2)$$

This is called Adams-Bashforth predictor formula.
Now we find

$$f_1 = f(x_0 + h, y_1)$$

Then to find a better value of y_1 , we derive a corrector formula by substituting Newton's Backward formula at f_1 i.e

$$f(x, y) = f_1 + n \nabla f_1 + \frac{n(n+1)}{2!} \nabla^2 f_1 + \frac{n(n+1)(n+2)}{3!} \nabla^3 f_1 \dots$$

in (1)

$$y_1 = y_0 + \int_{x_0}^{x_1} \left(f_1 + n \nabla f_1 + \frac{n(n+1)}{2!} \nabla^2 f_1 + \dots \right) dx$$

$$\text{Put } x = x_1 + nh \\ dx = h dn$$

$$= y_0 + \int_{-1}^0 \left(f_1 + n \nabla f_1 + \frac{n(n+1)}{2!} \nabla^2 f_1 + \dots \right) dn$$

$$= y_0 + h \left[f_1 - \frac{1}{2} \nabla f_1 - \frac{1}{12} \nabla^2 f_1 - \frac{1}{24} \nabla^3 f_1 - \dots \right]$$

Neglecting fourth & higher order differences & expressing ∇f_1 , $\nabla^2 f_1$ & $\nabla^3 f_1$ in terms of function values, we obtain

$$y_1 = y_0 + \frac{h}{24} [9f_1 + 19f_0 - 5f_{-1} + f_{-2}] \quad - (3)$$

which is called Adam-Moulton corrector formula.

Then an improved value of f_1 is calculated & again the corrector (3) is applied to find a still better value of y_1 . This step is repeated till y_1 remains unchanged & then we proceed to calculate y_2 as above.

eg: Solve the initial value problem $\frac{dy}{dx} = x - y^2$, $y(0) = 1$ to find $y(0.4)$ by Adam's method. Starting solutions required are to be obtained using Runge-Kutta method of order 4 using step value $h = 0.1$.

Solu: We have $f(x, y) = x - y^2$

To find $y(0.1)$:

$$x_0 = 0 \quad y_0 = 1 \quad h = 0.1$$

$$K_1 = hf(x_0, y_0) = (0.1)f(0, 1) = -0.1$$

$$K_2 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}K_1) = (0.1)f(0.05, 0.95) = -0.08525$$

$$K_3 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}K_2) = (0.1)f(0.05, 0.9574) = -0.08666$$

$$K_4 = hf(x_0 + h, y_0 + K_3) = (0.1)f(0.1, 0.9137) = -0.07348$$

$$K = \frac{1}{24} [K_1 + 2K_2 + 2K_3 + K_4] = -0.08621$$

$$y(0.1) = y_1 = y_0 + K = 1 - 0.08621 = 0.9137$$

To find $y(0.2)$

Here $x_1 = 0.1$ $y_1 = 0.9137$, $h = 0.1$

$$K_1 = h f(x_1, y_1) = (0.1) f(0.1, 0.9137)$$

$$K_2 = h f(x_1 + \frac{1}{2}h, y_1 + \frac{1}{2}K_1) = (0.1) f(0.15, 0.8751)$$

$$K_3 = h f(x_1 + \frac{1}{2}h, y_1 + \frac{1}{2}K_2) = (0.1) f(0.15, 0.8809)$$

$$K_4 = h f(x_1 + h, y_1 + K_3) = (0.1) f(0.2, 0.8491)$$

$$K = \frac{1}{6}(K_1 + 2K_2 + 2K_3 + K_4)$$

$$y(0.2) = y_2 = y_1 + K = 0.8494.$$

To find $y(0.3)$

$$x_2 = 0.2 \quad y_2 = 0.8494 \quad h = 0.1$$

$$K_1 = h f(x_2, y_2) = (0.1) f(0.2, 0.8494)$$

$$\begin{aligned} K_2 &= h f(x_2 + \frac{1}{2}h, y_2 + \frac{1}{2}K_1) \\ &= (0.1) f(0.25, 0.8233) \end{aligned}$$

$$\begin{aligned} K_3 &= h f(x_2 + \frac{1}{2}h, y_2 + \frac{1}{2}K_2) \\ &= (0.1) f(0.25, 0.828) \end{aligned}$$

$$\begin{aligned} K_4 &= h f(x_2 + h, y_2 + K_3) \\ &= (0.1) f(0.3, 0.8058) \end{aligned}$$

$$K = \frac{1}{6}(K_1 + 2K_2 + 2K_3 + K_4)$$

$$y(0.3) = y_3 = y_2 + K = 0.8061$$

Now the starting values of Adam's methods are

$$x=0.0 \quad y_{-3}=1.000$$

$$x=0.1 \quad y_{-2}=0.9137$$

$$x=0.2 \quad y_{-1}=0.8494$$

$$x=0.3 \quad y_0=0.8061$$

$$\text{We find } f_{-3} = f(0.0, 1.0) = 0.0 - (1.0)^2 = -1.000$$

$$f_{-2} = f(0.1, 0.9137) = 0.1 - (0.9137)^2 = -0.7312$$

$$f_{-1} = f(0.2, 0.8494) = 0.2 - (0.8494)^2 = -0.5215$$

$$f_0 = f(0.3, 0.8061) = 0.3 - (0.8061)^2 = -0.3498$$

Using the predictor formula

$$y_1 = y_0 + \frac{h}{24} (55f_0 - 59f_{-1} + 37f_{-2} - 9f_{-3})$$

$$x=0.4 \quad y_1 = 0.8061 + \frac{0.1}{24} [55 \times (-0.3498) - 59(-0.5215) + 37(-0.7312) - 9(-1.00)]$$

$$= 0.7789$$

$$\therefore f(0.4, 0.7789) = f_1 = 0.4 - (0.7789)^2 = -0.2067$$

Using corrector

$$y_1 = y_0 + \frac{h}{24} (9f_1 + 19f_0 - 5f_{-1} + f_{-2})$$

$$= 0.8061 + \frac{0.1}{24} [9 \times (-0.2067) + 19(-0.3498) - 5(-0.5215) + (-0.7312)]$$

$$= 0.7785$$

$$\therefore y(0.4) = 0.7785$$

(correct upto 3 decimal places)